Finite parts of power series as termwise series

Claude, with Daniel Murfet

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Abstract

Astra #8 rank 4 (face half). For a power series $a(v) = \sum_j z_j v^j$ with $|z_j|\rho^j \le H$ and $0 < b < \rho$, the weighted finite part at any admissible order $J > \gamma$ is the termwise absolutely convergent series $\mathrm{FP}_\gamma(a) = \sum_j z_j\, q_\gamma(j)$ with $q_\gamma(j) = b^{j-\gamma}/(j-\gamma)$ ($\log b$ at $j = \gamma$) (`axisFinitePart_series`). The proof follows Astra's advice to avoid the $\varepsilon \to 0$ limit: the Taylor remainder is the tail $\sum_{j\ge J} z_j v^j$, the weighted tail integrates termwise to $z_j b^{j-\gamma}/(j-\gamma)$, dominated by a geometric series, and the Bochner integral commutes with the sum. Applied to the analytic faces, the canonical face coefficient functions become explicit series $U_{\alpha_i}(s) = k_1^{-1}\sum_j c_{ij}(s) q_{\gamma_i}(j)$ (`canonU_anaFaceU_series`, and symmetrically for V). Zero `sorry`/`axiom`.

1 The things formalised

```
theorem axisFinitePart_congr (  b :  ) (f g :   →  ) (fm :   →  ) (M :  )
    (h :   v   Ioc (0 :  ) b, f v = g v) :
    axisFinitePart   b f fm M = axisFinitePart   b g fm M
theorem axisFinitePart_series (  b   H :  ) (hb : 0 < b) (hb  : b <  ) (z :   →  )
    (hz :   j, |z j| *   ^ j   H) (J :  ) (hJ :   < J) :
    (Summable fun j :   => z j * axisPrim   b j)
      axisFinitePart   b (fun v =>  ' j :  , z j * v ^ j) z J
        =  ' j :  , z j * axisPrim   b j
theorem row_majorant ... (i :  ) (s :  ) (j :  ) : |c (i, j) s| *   ^ j   H s /   ^ i
theorem faceFPCoeff_anaFaceU_series (c :   ×   →   →  ) (b    :  ) (H :   →  )
    (hb : 0 < b) (hb  : b <  ) (hc :   ij s, |c ij s| *   ^ (ij.1 + ij.2)   H s)
    (k i J :  ) (hJ :   < J) (s :  ) :
    faceFPCoeff   b k (anaFaceU c b i) (fun m => c (i, m)) J s
      = 1 / (k :  ) *  ' j :  , c (i, j) s * axisPrim   b j
theorem canonU_anaFaceU_series (c :   ×   →   →  ) (b   :  ) (H :   →  ) (h  h  k  k  :  )
    (hb : 0 < b) (hb  : b <  ) (hk  : 0 < k )
    (hc :   ij s, |c ij s| *   ^ (ij.1 + ij.2)   H s) (i :  ) :
    canonU b h  h  k  k  (anaFaceU c b) (fun i j s => c (i, j) s) (uExp h  k  i)
      = fun s => 1 / (k  :  )
          *  ' j :  , c (i, j) s * axisPrim ((k  :  ) * uExp h  k  i - h  - 1) b j
```

2 Mathematics

Write $e_j = (j+J)-1-\gamma > -1$. On $(0,b]$ the weighted remainder is $v^{-1-\gamma}\sum_{j\ge J} z_j v^j = \sum_j z_{j+J} v^{e_j}$ (tail of an absolutely convergent series, then $v^{-1-\gamma}v^{j+J} = v^{e_j}$). Each term is integrable with $\int_0^b z_{j+J} v^{e_j}\,dv = z_{j+J} b^{(j+J)-\gamma}/((j+J)-\gamma) = z_{j+J} q_\gamma(j+J)$ (the resonant branch is excluded since

$j+J \geq J > \gamma$), and $\int_0^b |z_{j+J}| v^{e_j}\, dv \leq |z_{j+J}| b^{j+J}\, b^{-\gamma}/(J-\gamma) \leq \frac{Hb^{-\gamma}}{J-\gamma}(b/\rho)^{j+J}$, a geometric majorant. Hence the regularised integral is the tail series $\sum_j z_{j+J} q_\gamma(j+J)$, which added to the counterterms $\sum_{j<J} z_j q_\gamma(j)$ gives the full series; summability of the full sequence follows from summability of its tail. For the analytic faces, $a_i(v,s) = \sum_j c_{ij}(s) v^j$ on $(0,b]$ (the clamp is the identity there), and the row majorant $|c_{ij}(s)|\rho^j \leq H(s)/\rho^i$ comes from the double-series envelope.

3 Paper-facing meaning

The paper's coefficients are "absolutely convergent series" in the Taylor data of ξ, η; our canonical constant coefficient $B_\alpha = M[\alpha;0;U_\alpha] + M[\alpha;0;V_\alpha] - M[\alpha;1;C_\alpha]$ now has $U_{\alpha_i}(s) = k_1^{-1} \sum_j c_{ij}(s)\, b^{j-\gamma_i}/(j-\gamma_i)$ (with $\log b$ at the resonant j), i.e. the finite part is the bounded linear functional $z \mapsto \sum_j z_j q_\gamma(j)$ on the weighted-ℓ^1 row. The remaining half of rank 4 is to pull the j-sum through the Gaussian s-moment ($M[\alpha;0;U_{\alpha_i}] = k_1^{-1} \sum_j q_{\gamma_i}(j) M[\alpha;0;c_{ij}]$), which needs the s-envelope $|c_{ij}(s)| \leq H(s)\rho^{-i-j}$ with $H(s) = Ye^{\beta s L}$ and the moment integrability already available.

4 Lean notes

- `Summable.sum_add_tsum_nat_add J` rewritten right-to-left, then `add_sub_cancel_left`, isolates the tail of a series in one step; `summable_nat_add_iff` transfers summability from the tail.
- `hasSum_integral_of_summable_integral_norm` takes integrability with respect to `volume.restrict (Ioc 0 b)`, which is what `IntegrableOn.const_mul` of `intervalIntegrable_rpow'` produces; its conclusion's integral prints as the set integral, so `← h.tsum_eq` fires directly.
- `gcongr` cannot discharge side goals like $0 \leq H$ for a bare variable; use `mul_le_mul` with explicit nonnegativity proofs.
- `rw [pow_add]` may leave a commutativity goal (`ring`) or none, depending on the association; `by rw [pow_add]; ring` is the safe form.
- `setIntegral_congr_fun` goals here needed no `beta_reduce` (the linter flags it as unused when the redex is absent).